
Overview

Citizen RAG Chatbot is an AI-powered assistant that helps citizens discover government welfare schemes, scholarships, pensions, healthcare benefits, and loan programs through natural language queries.

The system combines:

- **Retrieval-Augmented Generation (RAG)**
- **Hybrid Search (FAISS + BM25)**
- **Metadata Filtering**
- **Eligibility Engine**
- **Explainable AI (XAI)**
- **Large Language Models (Google Gemini)**

Instead of manually searching through hundreds of government portals, users can simply ask:

"I am a widow of an ex-serviceman."

or

"Show scholarship schemes for students."

and receive personalized scheme recommendations with explanations.

Features

Hybrid Retrieval System

- Semantic Search using **FAISS**
- Keyword Search using **BM25**

- Hybrid ranking and merging.
-

✓ Metadata Filtering

Filters schemes based on:

- Student
 - Widow
 - Ex-serviceman
 - Entrepreneur
 - Disability
 - Women
 - Senior Citizen
 - State-specific schemes
-

✓ Eligibility Engine

Automatically extracts user profiles and checks eligibility.

Example:

Input:

I am a widow of an ex-serviceman.

Profile:

```
{  
  "widow": true,  
  "ex_serviceman": true  
}
```

✓ Explainable RAG

Shows:

- Similarity Score
- BM25 Score
- Retrieval Method
- Why a scheme was retrieved

Example:

Why Retrieved:

- ✓ Matched query keywords
 - ✓ High semantic similarity
 - ✓ Matched metadata filter
-

✓ Scheme Explorer

Provides:

- Description
 - Benefits
 - Eligibility
 - Application Process
 - Official Scheme Details
-

✓ Follow-up Conversations

Example:

User:

Tell me about PMAY.

User:

What are the benefits?

User:

Who can apply?

The chatbot maintains scheme context during the conversation.

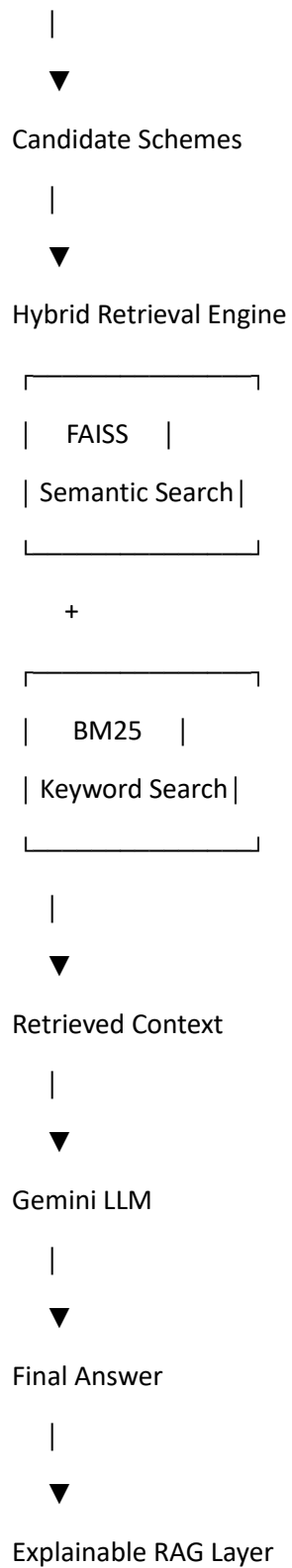
🔧 System Architecture

User Query

|



Metadata Filter



 Tech Stack

Technology

Purpose

Technology	Purpose
Python	Backend
Streamlit	Web Interface
FAISS	Vector Database
Sentence Transformers	Embedding Generation
BM25	Keyword Retrieval
Google Gemini	LLM
NumPy	Numerical Operations
JSON	Scheme Storage

Project Structure

Citizen-RAG-Chatbot/

```
|
|  └─ app.py
|  └─ eligibility.py
|  └─ metadata_filter.py
|  └─ bm25_retriever.py
|  └─ hybrid_retriever.py
|
|  └─ data/
|
|  └─ chunks.json
|  └─ enriched_schemes.json
|  └─ faiss_index.index
|  └─ embeddings.npy
|
|  └─ scripts/
|
|  └─ build_documents.py
|  └─ chunk_documents.py
|  └─ enrich_schemes.py
```

```
| |— generate_embeddings.py
|
|— requirements.txt
└— README.md
```

Installation

Clone Repository

```
git clone https://github.com/yourusername/Citizen-RAG-Chatbot.git
cd Citizen-RAG-Chatbot
```

Create Virtual Environment

```
python -m venv .venv
```

Windows

```
.venv\Scripts\activate
```

Linux/Mac

```
source .venv/bin/activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Run Application

```
streamlit run app.py
```

Example Queries

Scholarships

Show scholarship schemes for students

Medical Assistance

Medical assistance for ex-servicemen

Entrepreneurship

Loan schemes for entrepreneurs

Eligibility Queries

I am a widow of an ex-serviceman

I am a disabled student from Maharashtra

Retrieval Example

Scheme:

Skill Loan Scheme

Method:

Hybrid

Similarity Score:

0.84

BM25 Score:

7.12

Why Retrieved:

- ✓ Matched query keywords
- ✓ High semantic similarity
- ✓ Matched metadata filter

Applications

- Digital Governance
- E-Governance Portals
- Citizen Service Platforms
- Welfare Recommendation Systems
- Government Chatbots
- AI Assistants for Public Services

Future Improvements

- Multilingual Support

- Voice Assistant
 - Fine-Tuned LLM
 - OCR Document Upload
 - Personalized User Profiles
 - Scheme Recommendation Ranking
 - Mobile Application
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Author

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B.Tech CSE (AI & ML)

Interests:

- Artificial Intelligence
 - Machine Learning
 - Deep Learning
 - Generative AI
 - RAG Systems
 - DRDO & ISRO Research
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★ If you found this project useful, please consider giving it a star.